LETTERS

Mechanism of the ATP-dependent DNA end-resection machinery from *Saccharomyces cerevisiae*

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If not properly processed and repaired, DNA double-strand breaks (DSBs) can give rise to deleterious chromosome rearrangements, which could ultimately lead to the tumour phenotype[1,2]. DSB ends are resected in a 5′ to 3′ fashion in cells, to yield single-stranded DNA (ssDNA) for the recruitment of factors critical for DNA damage checkpoint activation and repair by homologous recombination[2]. The resection process involves redundant pathways consisting of nucleases, DNA helicases and associated proteins[3]. Being guided by recent genetic studies[4–6], we have reconstituted the first eukaryotic ATP-dependent DNA end-resection machinery comprising the *Saccharomyces cerevisiae* Mre11–Rad50–Xrs2 (MRX) complex, the Sgs1–Top3–Rmi1 complex, Dna2 protein and the heterotrimeric ssDNA-binding protein RPA. Here we show that DNA strand separation during end resection is mediated by the Sgs1 helicase function, in a manner that is enhanced by Top3–Rmi1 and MRX. In congruence with genetic observations[6], although the Dna2 nuclease activity is critical for resection, the Mre11 nuclease activity is dispensable. By examining the *top3 Y356F* allele and its encoded protein, we provide evidence that the topoisomerase activity of Top3, although critical for the suppression of crossover recombination[2,7], is not needed for resection either in cells or in the reconstituted system. Our results also unveil a multifaceted role of RPA, in the sequestration of ssDNA generated by DNA unwinding, enhancement of 5′ strand incision, and protection of the 3′ strand. Our reconstituted system should serve as a useful model for delineating the mechanistic intricacy of the DNA break resection process in eukaryotes.

The 3′-ssDNA strands derived from DSB resection attract RPA, which promotes the recruitment of checkpoint proteins to effect cell-cycle arrest[8]. With the aid of a recombination mediator protein, such as yeast Rad52 or human BRCA2, the Rad51 recombinase displaces RPA from the ssDNA to assemble into a right-handed helical polymer capable of initiating DSB repair by homologous recombination[1,2]. Genetic studies in yeast have shown that DSB resection proceeds in two steps. The MRX complex plays a role in initiation, whereas the Sgs1 helicase, its associated proteins Top3 and Rmi1, and the helicase/nuclease Dna2, whose nuclease activity is needed for Okazaki fragment processing[9,10], constitute the DNA motor-driven path of long-range resection. Exo1, a 5′ to 3′ exonuclease, defines a redundant resection means[4–6]. Here we reconstitute the Sgs1/Dna2-dependent DNA resection machinery and present results germane for understanding its mechanistic underpinnings.

The requisite factors, namely Sgs1, Top3–Rmi1 complex, MRX complex, Dna2 and RPA were purified and analysed (see Supplementary Fig. 2 and Supplementary Information). As shown in Fig. 1A, the combination of these factors rapidly degraded a 1.9-kilobase (kb), linear, uniformly ^{32}P-labelled duplex. Little or no digestion occurred when Dna2, Sgs1 or RPA were omitted. In the absence of Dna2, the DNA substrate was unwound completely to yield ssDNA, which was not seen upon omission of Sgs1, indicating that the Sgs1 helicase activity but not that of Dna2 can support complete unwinding. The results also revealed a dependence of DNA unwinding on RPA. In the absence of MRX, DNA degradation was attenuated, a result similar to the delayed resection observed in *mre11Δ* cells[3,6,11]. Also in concordance with genetic results[6], Top3–Rmi1 stimulated

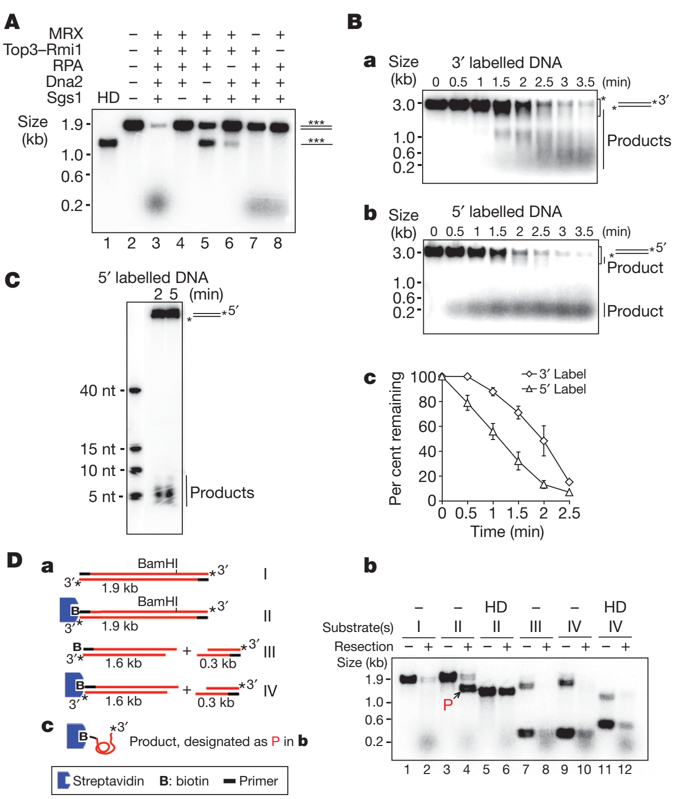

Figure 1 | Reconstitution and 5′ polarity of DNA end resection. A, Resection of uniformly labelled DNA. HD, heat denaturation of DNA. **B,** Time-course analysis of resection with 3′-labelled (a) or 5′-labelled (b) DNA. Mean values ± s.d. from three independent experiments were plotted (c). The asterisks in (a) and (b) mark the area that was quantified. **C,** Analysis of the initial 5′ resection product in a denaturing polyacrylamide gel; nt, nucleotides. **D,** The biotinylated substrates (a) were incubated with the resection machinery for 5 min as in (**A**) and analysed (b). Complete digestion of the unblocked 5′ strand yielded ssDNA (designated P in (b) and depicted in (c)) that harboured biotin-streptavidin at the 5′ terminus and ^{32}P at the 3′ terminus.

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resection. The ability to degrade DNA required ATP, which could not be replaced by ATP-γ-S or AMP–PNP (Supplementary Fig. 3a). Thus ATP hydrolysis is needed for DNA degradation. The stimulatory effects of the MRX and Top3–Rmi1 complexes are addressed in greater detail later.

Importantly, supercoiled DNA was not degraded by the same combination of protein factors (Supplementary Fig. 3b), confirming that the reconstituted resection machinery is DNA-end specific. The top3 Y356 mutant that lacks topoisomerase activity was used in this experiment (Supplementary Fig. 3b, c) to avoid the relaxation of the supercoiled DNA. As detailed later, the *top3-Y356F* mutant is differentially inactivated for the ability to attenuate mitotic crossover recombination.

We used 3-kb 3′- and 5′-labelled linear duplex DNA substrates to test the polarity of resection. The 3′-labelled DNA was converted to intermediates that progressively decreased in size over the course of the reaction (Fig. 1B). In contrast, much of the product generated from 5′-labelled DNA migrated at the front of the agarose gel (Fig. 1B). The radiolabelled product derived from the 5′-labelled substrate ranged in size from four to eight nucleotides, with the five-nucleotide size being predominant (Fig. 1C). Incubation with 3′-labelled DNA did not generate any small product until much later (Supplementary Fig. 4b). To determine whether conversion of the 3′ label into the small product stemmed from progressive digestion of the 5′ strands or from limited resection of the 3′ strands, we designed a 3′ ^{32}P-labelled substrate in which one end was tagged with biotin and could therefore be blocked from protein access with streptavidin. The results showed that a large fraction of the 3′ ^{32}P-label at the unblocked end remained intact during resection (Fig. 1D). Together, the results show that the reconstituted DNA end-resection machinery has a strong specificity for the 5′ strand and incises the 3′ strand rather infrequently.

Dna2 possesses an endonuclease activity capable of digesting 3′ or 5′ ssDNA[12], whereas Mre11 harbours a 3′ to 5′ exonuclease activity and a DNA-structure-specific endonuclease activity[13–15]. Using the nuclease null dna2-D657A protein (Supplementary Fig. 5) and mre11-3 protein[16] (Supplementary Fig. 6) with the 3′-labelled substrate, we found that extensive resection is mediated by Dna2 nuclease (Fig. 2A). Even when manganese ion was included to activate the Mre11 nuclease (Supplementary Figs 6d and 7)[13,17], 5′ cleavage was still completely dependent on Dna2 (Supplementary Fig. 7b, c). Although Sae2 protein has been implicated in the initial step of end resection[5,6], its addition did not affect the efficiency of the first cleavage or product size (Supplementary Fig. 8).

The dependence of DNA resection on ATP hydrolysis (Supplementary Fig. 3a, b) implies that either or both of the DNA motor functions residing in Sgs1 and Dna2 might be indispensable. We therefore purified the ATPase-defective variants, dna2-K1080E[9] and sgs1-K706A[18] (Supplementary Figs 5 and 9), and tested them with 3′-labelled DNA. Substitution of Sgs1 with sgs1-K706A led to ablation of resection, but resection occurred normally with dna2-K1080E replacing Dna2 (Fig. 2B). We note that these biochemical results are consistent with genetic observations that the Sgs1 helicase activity, but not the Dna2 helicase activity, is indispensable for DSB resection[5,6].

We investigated the specificity of some of the protein factors by replacing Sgs1 with the Srs2 helicase that functions in DNA damage repair and response[1] and Dna2 with Fen1 (product of the *RAD27* gene), a 5′ FLAP endonuclease involved in Okazaki fragment processing[19]. We found that Srs2 and Fen1 are non-functional in resection (Fig. 2C). These biochemical results are in congruence with genetic data showing that deletion of *SRS2* has no impact on the rate and extent of DSB resection[6], and that *RAD27* fails to suppress the resection defect of *pif1-m2 dna2Δ* cells (Fig. 2D). Our results also suggest that suppression of the *dna2* mutant phenotype by *RAD27* overexpression[20] stems from the restoration of Okazaki fragment processing.

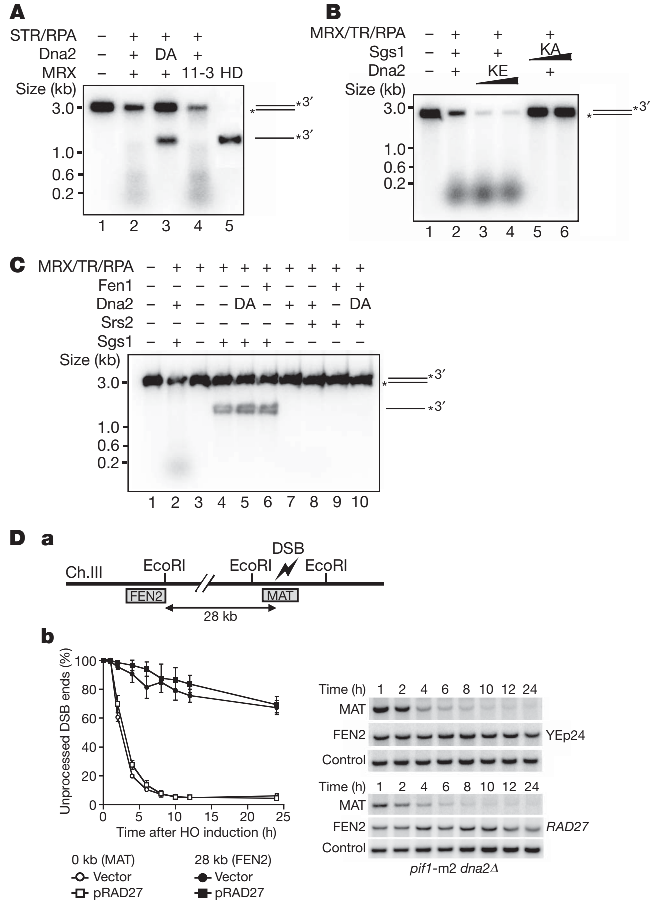

Figure 2 | Requirement for Sgs1 helicase and Dna2 nuclease activities. **A**, The dna2-D657A (DA) mutant and mutant MRX harbouring mre11-3 (11-3) were tested with 3′-labelled substrate. STR, Sgs1–Top3–Rmi1. **B**, The dna2-K1080E (KE) and sgs1-K706A (KA) mutants were tested. **C**, Srs2 helicase, Fen1 nuclease or their combination failed to substitute for Sgs1 and Dna2. DA, dna2-D657A. **D**, *RAD27* overexpression (YEp24) failed to suppress the end-resection defect of *dna2Δ* cells. Initial resection next to the break (*MAT* probe) and long-range resection (*FEN2* probe) were analysed[6], as depicted (a). A set of primary data and the mean values ± s.d. from three independent experiments are presented (b).

The effects of MRX and Top3–Rmi1 on resection were further defined with a limiting amount of Dna2 and Sgs1 and uniformly labelled DNA. Although the combination of Sgs1/Dna2/RPA was able to mediate resection, the addition of MRX or Top3–Rmi1 stimulated the reaction by about fourfold or eightfold, respectively (Fig. 3A). Consistent with results shown earlier (Fig. 1A), the most efficient resection was seen when both MRX and Top3–Rmi1 were included (Fig. 3A). Interestingly, MRX or Top3–Rmi1 alone could stimulate the Sgs1-mediated unwinding of a 3-kb substrate, by about fourfold or eightfold, respectively, and an additive effect was seen (Fig. 3C). The results from *in vitro* pulldown revealed that MRX interacts with Sgs1 at physiological ionic strength (Fig. 3B). This agrees with a previous study showing co-fractionation and co-immunoprecipitation of Sgs1 with Mre11 (ref. 21). MRX also associates with Dna2 (Fig. 3B), although it does not affect the activity of Dna2 nuclease (Supplementary Fig. 10c). Because MRX binds DNA ends[22], it could be involved in the DNA end recruitment of Sgs1 and Dna2. This premise is supported by the finding that DNA that had been pre-processed to generate 3′-ssDNA tails could be resected efficiently by Sgs1–Dna2–RPA with or without MRX (Supplementary Fig. 11). We also verified that the Top3–Rmi1 complex binds Sgs1 (ref. 23) and noted a weak interaction between Top3–Rmi1 and Dna2 as well (Fig. 3B). We did not detect any significant physical interaction of Top3–Rmi1 with MRX and, using protection of 5′-labelled DNA against lambda exonuclease[22]

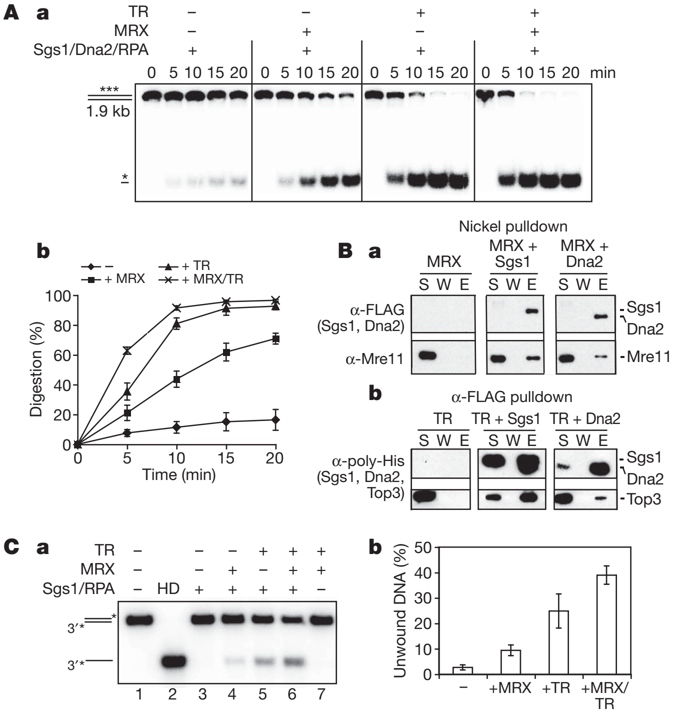

Figure 3 | Role of MRX and Top3–Rmi1 in DNA resection and unwinding. **A**, The resection role of MRX and Top3–Rmi1 was analysed with uniformly labelled DNA. Reactions were resolved in 10% polyacrylamide gels (a), and the mean values ± s.d. from three independent experiments were plotted (b). **B**, Pulldown through the His_6 tag (Nickel) or FLAG tag (α-FLAG) on FLAG-His_6-tagged Dna2 or Sgs1 to test for their interaction with MRX (a) or His_6-Top3–Rmi1 complex (b). The supernatant (S), wash (W) and SDS eluate (E) fractions were immunoblotted with the indicated antibodies. **C**, Sgs1 and RPA were examined for the unwinding of a 3-kb DNA in the absence or presence of MRX and/or Top3–Rmi1 (a), and the mean values ± s.d. from three independent experiments were plotted (b).

as assay, found no specific affinity of Top3–Rmi1 for DNA ends (data not shown).

Top3 cooperates with Rmi1 and Sgs1 in the suppression of mitotic crossover formation[7], presumably by dissolution of the double Holliday junction[24]. Results shown earlier (Supplementary Fig. 3b) suggested that the catalytically null top3-Y356F mutant could function in resection. Indeed, a direct comparison with wild-type Top3 revealed no deficiency in the top3-Y356F mutant in resection (Fig. 4A). Likewise, *top3-Y356F* mutant cells are proficient in resecting the HO-induced DSB (Fig. 4B). However, although double Holliday junction dissolution occurs efficiently with Sgs1, Rmi1 and Top3 *in vitro*, top3-Y356F is completely defective in this regard (Supplementary Fig. 3d). Consistent with the biochemical data, the *top3-Y356F* allele is unable to suppress the high level of mitotic crossovers in *top3Δ* cells (Fig. 4C). These results indicate that the catalytic activity of Top3 is critical for crossover control but dispensable for DSB resection.

Earlier we saw a strict dependence of DNA resection on RPA (Fig. 1A), which is expected to facilitate the DNA unwinding step by sequestering ssDNA generated by Sgs1. Our finding of DNA unwinding in the absence of Dna2 as being strongly reliant upon RPA (data not shown) and the fact that human RPA (hRPA) or *Escherichia coli* SSB is able to support DNA unwinding (Fig. 4D) are consistent with this premise. However, little or no resection occurred with either hRPA or SSB (Fig. 4D), indicating that yeast RPA (yRPA) must fulfil another role in resection. That yRPA, but neither hRPA nor SSB, physically interacts with Dna2 reinforces this idea[10] (Supplementary Fig. 10a). Alternatively, or in addition, hRPA and SSB may inhibit the nuclease activity of yeast Dna2. To understand the specificity of yRPA in resection, we used partly duplex DNA substrates that bore either a labelled 3′- or 5′-ssDNA tail to examine

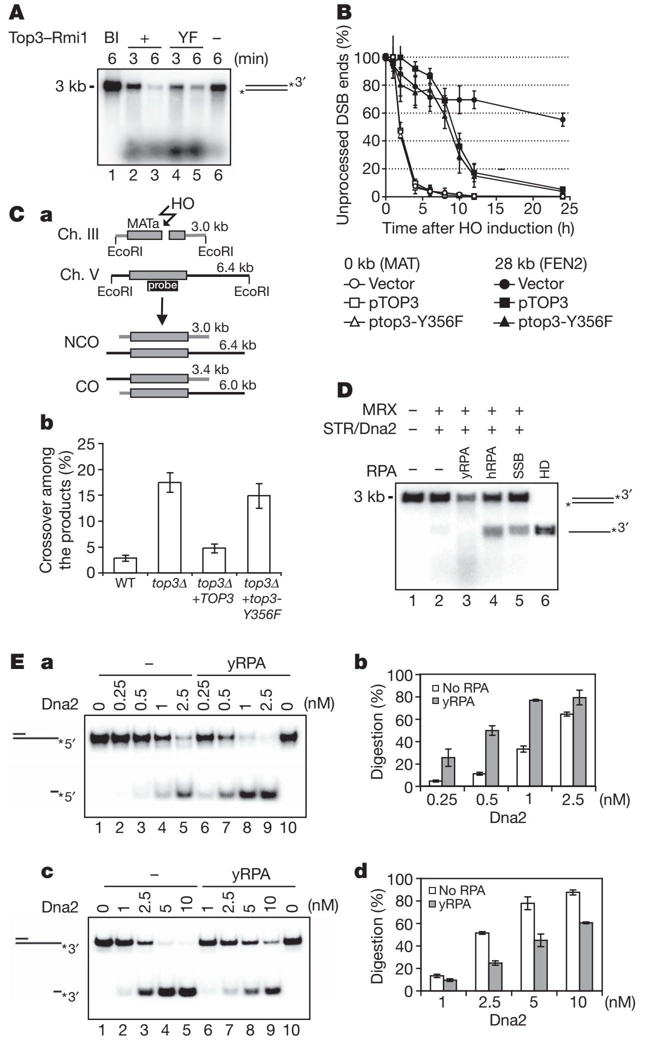

Figure 4 | Role of Top3 and RPA in resection. **A**, Top3–Rmi1 and top3-Y356F-Rmi1 were tested in the resection system with 3′-labelled substrate. Bl, no protein added. **B**, Analysis of 5′ resection in *top3Δ* cells supplemented with the low copy pRS315 vector carrying the *TOP3* or *top3-Y356F* gene. Analysis was done as in Fig. 2D. **C**, Crossover (CO) and non-crossover (NCO) recombinants are distinguished based on their size (a). The frequencies of crossover recombinants 8 h after break induction were determined[7] and the mean values ± s.d. from three independent experiments were plotted (b). **D**, SSB or hRPA supports DNA unwinding but not end resection. **E**, Yeast RPA stimulates 5′ end digestion (a) but attenuates 3′ end digestion (c) by Dna2. The mean values ± s.d. from three independent experiments were plotted in (b) and (d).

how yRPA, hRPA and SSB affect the nuclease attribute of Dna2. Interestingly, yRPA enhanced 5′-DNA incision by Dna2 but, in reproducible fashion, attenuated the 3′-DNA cleavage reaction (Fig. 4E). We note that yRPA was previously found to reduce the cleavage of the 3′ tail of a substrate bearing a G4 DNA structure[25]. Importantly, strong inhibition of the Dna2 nuclease function by hRPA and SSB was seen (Supplementary Fig. 10b). The dna2-Δ405N mutant protein, which is partly defective in RPA interaction and Dna2 nuclease enhancement[26] (Supplementary Fig. 12a, b), is stimulated by yRPA to a lesser degree (Supplementary Fig. 12c) and, accordingly, is significantly less effective than Dna2 in DNA resection (Supplementary Fig. 12d).

Our studies have clarified the role of the Sgs1 helicase and Dna2 nuclease activities and of the MRX and Top3–Rmi1 complexes in DSB resection in a reconstituted system. RPA is found to be an essential component of the Sgs1/Dna2 resection pathway, by supporting the DNA unwinding step and enhancing the 5′-endonuclease activity of Dna2. Importantly, RPA attenuates 3′-DNA cleavage by Dna2, an attribute that likely serves to enforce the 5′ strand specificity of the resection machinery. The mechanistic details of the DNA resection machinery are depicted in our working model (Supplementary Fig. 1).

The reconstituted DNA motor-driven resection machinery described herein and in the accompanying paper by Cejka *et al.*[27] comprises ten polypeptides, which is much more complex than the resection systems from prokaryotes and archaea, such as *E. coli* RecBCD (three polypeptides), AdnAB (two polypeptides) from mycobacteria or Mre11–Rad50–HerA–NurA (four polypeptides) from *Pyrococcus furiosu*[28]. Moreover, the dependence on an ssDNA binding protein (that is, RPA) is not evident in these other systems. The process of DNA damage repair is linked to checkpoint-mediated cell-cycle control, wherein RPA-coated ssDNA activates the ATR/Mec1 kinase through ATRIP/Ddc2 (ref. 8). Based on our results, we propose that RPA plays an even earlier role in checkpoint activation by promoting resection. As such, RPA could provide a means for coupling the decision of DSB end processing with checkpoint activation. The reconstituted system should be beneficial for understanding the mechanism of DSB resection in humans, as there is evidence that the Sgs1 orthologue, BLM, which is mutated in the cancer-prone Bloom syndrome, is similarly involved in DSB processing[4]. It will be of interest to test whether human DNA2, found in both mitochondria and the nucleus[29,30], also helps mediate chromosomal end resection in a manner analogous to yeast Dna2.

METHODS SUMMARY

The proteins and protein complexes (that is, Mre11–Rad50–Xrs2, Dna2, yeast and human RPA proteins, Sgs1, Top3–Rmi1, Srs2, Fen1) used in this study were expressed in insect, yeast or *E. coli* cells and purified as detailed in Supplementary Information. *E. coli* SSB protein was purchased (New England BioLabs). For the end-resection reactions, linear DNA was 3′- or 5′-labelled with ^{32}P using standard methods, and the internally labelled [^{32}P]DNA was generated by PCR. Other [^{32}P]DNA substrates for testing Sgs1 and Dna2 DNA helicase and Dna2 nuclease activities were prepared as described in the Supplementary Information. The supercoiled ϕX174 DNA was purchased (New England BioLabs) and its linearization was by digestion with the restriction enzyme StuI. The DNA resection reactions were analysed in agarose or polyacrylamide gels under denaturing or non-denaturing conditions, followed by phosphorimaging analysis of the dried gel or ethidium bromide treatment of the gel to visualize DNA species. The pulldown experiments to test for protein–protein interactions made use of affinity tags on the indicated proteins. Visualization of proteins was by the staining of SDS polyacrylamide gels with Coomassie blue or by immunoblotting. Experimental details for the Dna2 nuclease, double Holliday junction dissolution, DNA helicase and ATPase assays, and pre-resection of uniformly [^{32}P]DNA can be found in Methods or Supplementary Information. Genetic assays to measure the distribution of recombinant products of the gene conversion and crossover types or the kinetics of the resection of a site-specific DNA double-strand break were conducted as described[6,7].

Full Methods and any associated references are available in the online version of the paper at www.nature.com/nature.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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Author Contributions H.Y.N, G.I. and P.S. designed the experiments and wrote the paper. H.Y.N., W.-H.C., Z.Z., Y.H.K., P.C., W.X.Z. and R.P. conducted the experiments. L.L. and D.L. provided key materials and technical expertise.

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METHODS

DNA end-resection substrates. The 3-kb pBluescript SK(-) was linearized with EcoRV and then 5′-labelled with [γ-^{32}P]ATP (PerkinElmer) and T4 polynucleotide kinase (New England Biolabs) or 3′-labelled with [α-^{32}P]cordycepin 5′-triphosphate (PerkinElmer) and terminal deoxytransferase (Roche). The 1.9-kb internally labelled substrate was generated by PCR in the presence of [α-^{32}P]dCTP (PerkinElmer) and using 5′-GCCGAGCCATATGTCAAGTGAG TCAACAACTTTCATCGTGG-3′ and 5′-GCACGACTCGAGTTAATTATTGC TATTGTTTGGACTTCCCC-3′ as primers and the *S. cerevisiae HDF2* gene as template. The substrates were purified from 1% agarose gels with the Gel Extraction Kit (Qiagen). The substrate in Fig. 1D was generated using the same template and primers, except with the first primer being biotinylated at the 5′ end. To block the DNA end that harboured the biotin label, a 10-μl reaction (50 mM Tris-HCl, pH 7.9, 100 mM NaCl, 5 mM $MgCl_2$, 1 mM DTT) containing substrate (5-nM ends) and 0.5 mg ml^{-1} streptavidin (Invitrogen) was incubated at 25 °C for 20 min. Where indicated, the substrate was treated with BamHI to yield two fragments (1.6 kb and 0.3 kb), the larger of which bore the biotin. Biotinylation of the 5′ terminus reduced the efficiency at which the companion 3′ terminus was labelled with ^{32}P (Fig. 1D, lane 7).

DNA end-resection assays. Reactions were conducted in 10-μl buffer (40 mM Tris-HCl, pH 7.5, 2 mM ATP, 4 mM $MgCl_2$, 50 mM KCl, 1 mM DTT, 100 μg ml^{-1} BSA, an ATP-regenerating system of 20 mM creatine phosphate and 20 μg ml^{-1} creatine kinase) and the specified DNA substrate (0.5 nM ends) at 30 °C. Unless specified otherwise, each reaction contained 10 nM MRX complex, 10 nM Sgs1, 10 nM Top3–Rmi1 complex, 20 nM Dna2 and 100 nM RPA. The reaction was deproteinized by SDS (0.2%) and proteinase K (0.5 mg ml^{-1}) at 37 °C for 10 min before analysis in an agarose gel (1 or 1.2%) in TAE (35 mM Tris-acetate, pH 7.4, 0.5 mM EDTA) buffer. Alternatively, reaction mixtures were resolved in a 10% polyacrylamide gel in TBE (89 mM Tris-borate, pH 8.0, 2 mM EDTA) buffer or an 18% polyacrylamide gel under denaturing (7 M urea in TBE buffer) conditions. Gels were dried onto DE3 paper (Whatman) before phosphorimaging analysis. For analysing the resection product size (Fig. 1C), the amount of DNA substrate was increased fivefold (2.5 nM ends). For time-course analysis testing MRX and Top3–Rmi1 effects (Fig. 3A), we set up 30-μl reactions containing Sgs1 (1 nM), Dna2 (1 nM), RPA (100 nM) without or with MRX (2 nM), Top3–Rmi1 (1 nM) or both MRX and Top3–Rmi1, and internally labelled DNA (0.13-nM DNA ends) as substrate.

Effect of MRX and Top3–Rmi1 on Sgs1-mediated DNA unwinding. The 3-kb 3′-labelled DNA substrate (0.5 nM ends) was incubated with Sgs1 (5 nM) and RPA (100 nM) in 10 μl buffer (30 mM Tris-HCl, pH 7.5, 1 mM DTT, 100 μg ml^{-1} BSA, 2 mM $MgCl_2$, 2 mM ATP and the ATP regenerating system) at 30 °C for 5 min. The reaction mixtures were deproteinized, resolved in a 1.2% agarose gel in TAE buffer, dried onto DE3 paper and subjected to phosphorimaging analysis.

Dna2 nuclease assay. Dna2 was incubated with or without yRPA, hRPA or SSB (20 nM each) at 4 °C for 5 min in 9.5 μl buffer (25 mM Tris-HCl, pH 7.5, 1 mM DTT, 100 μg ml^{-1} BSA, 100 mM KCl, 2 mM $MgCl_2$). After the addition of the indicated DNA substrate (5 nM), the completed reaction (10 μl) was incubated at 37 °C for 20 min. The reaction mixtures were deproteinized and resolved in a 10% polyacrylamide gel at 4 °C in TBE buffer, and the gel was dried onto DE3 paper before phosphorimaging analysis.

Affinity pulldown assays. To test for MRX–Sgs1 or MRX–Dna2 interaction, MRX (750 ng) was incubated with FLAG-His$_6$-tagged Sgs1 (250 ng) or Dna2 (200 ng) at 4 °C for 30 min in 30 μl buffer (10 mM Na_2HPO_4, 1.8 mM KH_2PO_4, pH 7.4, 137 mM NaCl, 3 mM KCl, 0.05% Igepal CA-630 (Sigma), 1 mM DTT, 20 mM imidazole). After mixing with 8 μl of nickel-NTA resin at 4 °C for 30 min, the resin was washed three times with 300 μl of buffer, and 20 μl of 2% SDS was used to elute proteins. The supernatant, last wash and SDS eluate fractions, 5 μl each, were resolved by 8% SDS–PAGE and immunoblotted with either anti-FLAG or anti-Mre11 antibodies.

To test for Top3–Rmi1–Sgs1 or Top3–Rmi1–Dna2 interaction, Top3–Rmi1 (100 ng) and Sgs1 (250 ng) or Dna2 (200 ng) were incubated as above, except that anti-Flag-M2 resin (Sigma) was used to capture protein complexes. Analysis was as above, with anti-poly-histidine antibodies being used to reveal Sgs1, Dna2 and Top3.